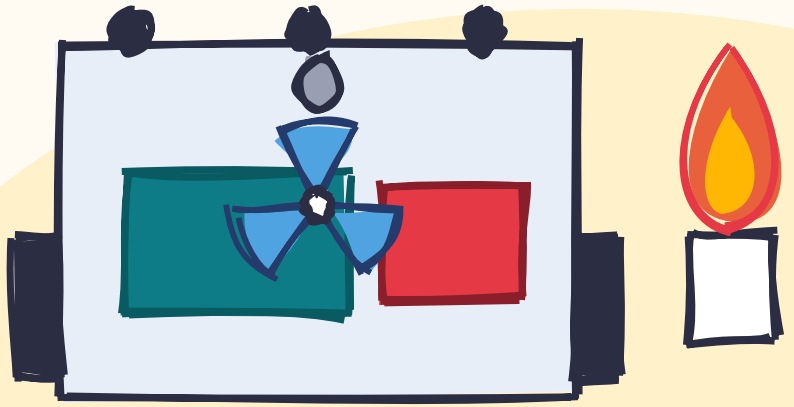
Tech Leaders · Class 14

Autonomous Fire-Fighting Robot

Find the flame, drive to it, put it out. No remote control.

Flame sensors | Steering | State machines | Relay + fan



Did you know?
Fire-fighting robots helped fight the 2019 Notre-Dame fire in Paris, going where it was too dangerous for people.

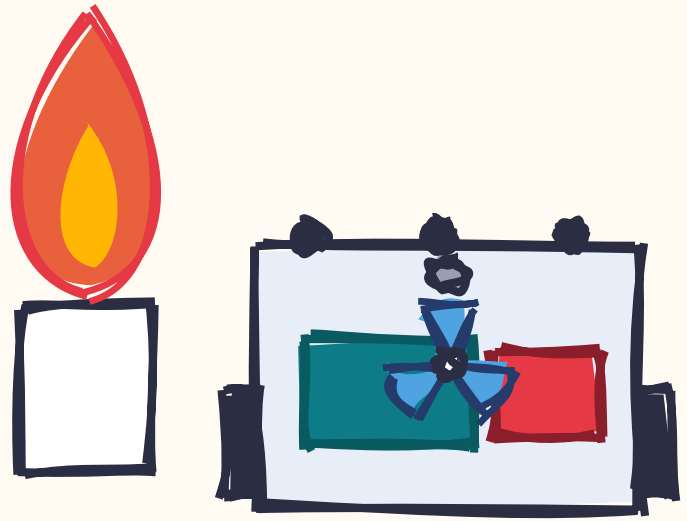
LEARN

Let's think...

Would you send a person, or a robot?

A burning warehouse

A chemical warehouse is on fire. The roof might collapse. Firefighters need to get water to the flames, but going inside could kill them. Robots can't feel heat or fear. But how would a robot **find** a fire and know what to do, all by itself?



Remote control?

A camera?

Sensors + logic!

Today: you build a robot that **searches** for a candle, **drives** to it and **blows it out**, with no human control.

Learning objectives

By the end of this class I can...

1	**Sense flame:** read 3 **IR flame sensors** and calibrate 'seen' and 'close' thresholds	**Proof:** our table of readings at 1 m, 50 cm and 20 cm
2	**Steer:** steer towards the **strongest** sensor reading	**Proof:** the robot turns towards the candle from any side
3	**State machine:** organise behaviour as a **state machine**: search, approach, extinguish	**Proof:** Serial Monitor prints each state change
4	**Actuate safely:** switch a fan with a **relay** and sweep it with a servo	**Proof:** the fan sweeps and the flame goes out
5	**Test & evaluate:** run timed trials, find **false alarms** and improve	**Proof:** results from 5 trials and one fix

LEARN

Objective 4 · Actuate safely · video

Video time!



Explosion-Proof Fire Fighting Robot Field Test

Wei's Technology · about 3 minutes · youtu.be/gH4rPVHxfhA

Watch for...

1. How does the robot **move** over rough ground?
2. How far away can it stay from the fire while spraying?
3. Is this robot **autonomous** or remote-controlled? How can you tell?

Scan to watch

After the video: List 3 jobs in this video a robot did better than a human, and 1 job a human still had to do.

LEARN

Objective 1 · Sense flame

Seeing fire with infrared

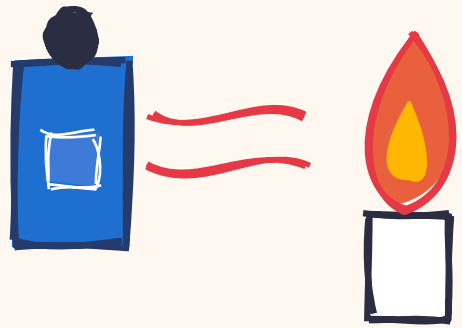
The idea

Flames give off **infrared (IR)** light, invisible to us. The flame sensor's dark bulb is an IR **photodiode**, tuned to about 760–1100 nm.

On most modules the **analog output falls** as IR gets stronger: about 1000 = nothing, 500 = flame across the room, 100 = very close. **Calibrate yours!**

False alarms
Sunlight and halogen lamps contain lots of IR too. Test away from windows, or the robot chases the sun!

Example: seeing heat with a thermal camera



Firefighters use thermal cameras to see hot spots through smoke. Our sensor is a one-pixel version: it can't make a picture, but it can tell 'how much IR is reaching me'.

Objective 2 · Steer

Turn towards the strongest signal

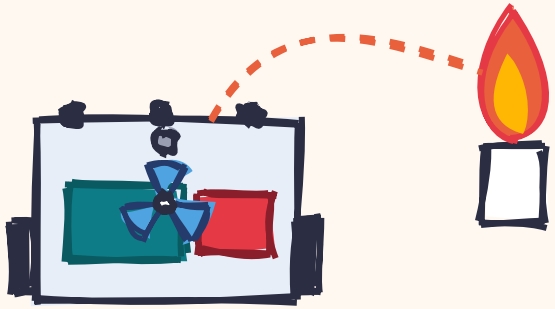
The idea

Read L, C and R. The **smallest** number is the strongest IR.

- Strongest on **L** → slow the left wheel, speed the right: the robot curves left.
- Strongest on **R** → the opposite.
- Strongest on **C** → both wheels forward.

Done many times a second, the robot homes in on the flame like a moth.

Example: a moth to a lamp



A moth flies towards light by keeping it in a particular place in its eyes and correcting constantly. Small, fast corrections add up to a smooth path.

Objective 3 · State machine

A state machine keeps the robot organised

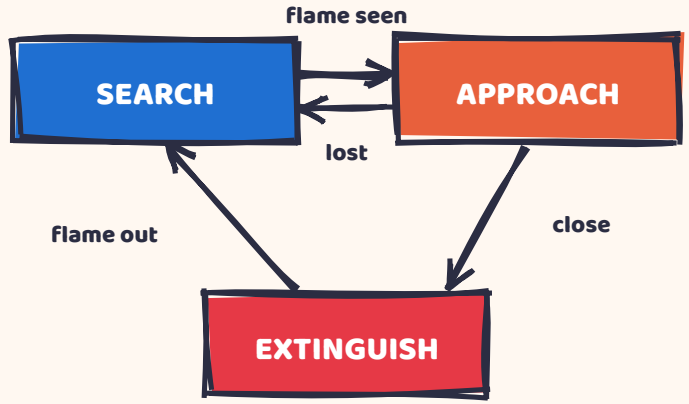
The idea

The robot is always in exactly **one state**, with clear rules to switch:

- **SEARCH**: spin slowly. Flame seen → APPROACH.
- **APPROACH**: steer towards it. Lost → SEARCH. Close → EXTINGUISH.
- **EXTINGUISH**: stop, fan on, sweep. Flame gone → SEARCH.

Without states, code becomes a tangle of if-statements fighting each other.

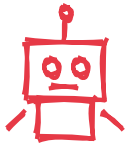
Example: traffic lights



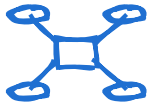
A traffic light is only ever green, amber or red, and it changes in a fixed order on clear triggers (a timer, a button). Our robot's 'colours' are search, approach and extinguish.

Real world

Robots on the front line



Fire robots
Tracked robots spray water inside burning buildings



Drones
Thermal cameras find hot spots in forest fires



Bomb disposal
Remote robots handle explosives



Disaster zones
Robots searched Fukushima's damaged reactors

? Should a fully autonomous robot ever decide to enter a building where people might be trapped?

Human in charge
Today, most rescue robots are 'teleoperated': a human makes the big decisions from a safe distance.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Flame rules for this lab
2. Hot and cold
3. Our robot kit (per group)
4. Wire the robot
5. Calibrate the eyes
6. Program part 1: setup and drive()
7. Program part 2: the state machine

Safety first

Flame rules for this lab

Teacher only lights and puts out the candle. Tealight in a metal tray on sand.

Clear zone no paper, bags, hair or sleeves near the arena. Long hair tied back.

Always watched a burning candle is never left alone. Water and a fire blanket ready.

Robot stops short it must stop at least 15 cm from the flame. It blows; it never touches.

Low voltage fan and motors run on batteries only.



Test cold first

Test steering with a TV remote or phone torch first. Real flame only for final trials.

Class game

Hot and cold

How to play

1. One student is the robot, eyes closed, with two 'sensor' helpers on their left and right.
2. The 'flame' is a student holding a ringing phone somewhere in the room.
3. Helpers may only say how loud it is on their side: **1 to 5**.
4. The robot turns towards the louder side and steps forward.
5. Stop one step away and 'blow it out'!

Then discuss

What rule did the robot follow? What happened when both sides said the same number?



Big idea

Compare left and right, turn towards the bigger signal. That is all our robot needs.

Build it

Our robot kit (per group)



Arduino Uno
The robot's brain



3 flame sensors
Left, centre, right eyes



2WD chassis + L298N
Movement



Relay + 5 V fan
The 'extinguisher'



Servo SG90
Sweeps the fan side to side



Buzzer + LED
Alarm when fire found

Also: 4×AA pack, jumper wires, a teacher-controlled tealight in a metal tray.

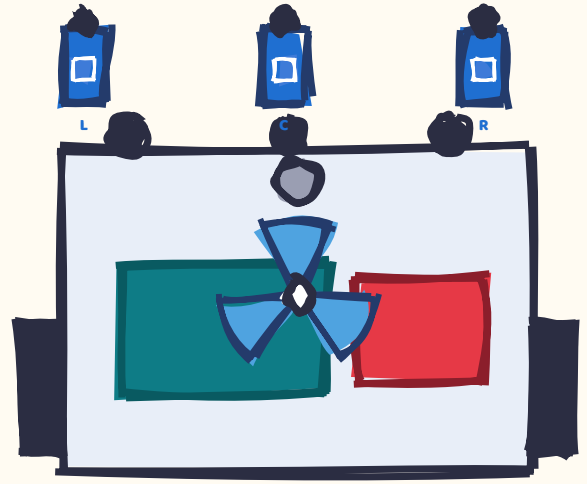
Why a fan?
Real robots use water or foam. A fan is safe, dry and still has to be aimed precisely.

MAKE

Objective 1 · Sense flame · Build step 1 of 4

Wire the robot

Part	Connect to
Flame sensors AO (L, C, R)	A0, A1, A2
Flame sensors VCC / GND	5V / GND
L298N ENA, IN1, IN2	5, 7, 8
L298N IN3, IN4, ENB	9, 10, 6
Servo signal	pin 3
Relay IN (fan via COM/NO)	pin 4
Buzzer / LED	11 / 12



Angle the eyes

Point L and R about **30–40°** outwards so they see different directions. Overlapping eyes can't tell left from right.

Objective 1 · Sense flame · Build step 2 of 4

Calibrate the eyes

```
void setup() { Serial.begin(9600); }

void loop() {
  Serial.print(analogRead(A0)); Serial.print("\t");
  Serial.print(analogRead(A1)); Serial.print("\t");
  Serial.println(analogRead(A2));
  delay(200);
}
```

Test	L	C	R
No flame			
Candle at 1 m			
Candle at 50 cm			
Candle at 20 cm			
Phone torch at 20 cm			

Do it twice

Once with the lights on, once near the window. Did the 'nothing' reading change?

Your thresholds

SEE = a bit below 'no flame'. CLOSE = your 20 cm reading.

Objective 3 · State machine · Build step 3 of 4

Program part 1: setup and drive()

```
#include <Servo.h>
Servo sweep;
const int ENA=5, IN1=7, IN2=8, IN3=9, IN4=10, ENB=6;
const int FAN = 4, BUZ = 11;
const int SEE = 900, CLOSE = 250;      // from YOUR calibration

enum State { SEARCH, APPROACH, EXTINGUISH };
State state = SEARCH;

void drive(int l, int r) {             // -255 .. 255 per wheel
  digitalWrite(IN1, l > 0);  digitalWrite(IN2, l < 0);
  analogWrite(ENA, abs(l));
  digitalWrite(IN3, r > 0);  digitalWrite(IN4, r < 0);
  analogWrite(ENB, abs(r));
}

void setup() {
  for (int p = 4; p <= 12; p++) pinMode(p, OUTPUT);
  sweep.attach(3);
  Serial.begin(9600);
}
```

enum

Gives the states readable names. **state == SEARCH** is clearer than **state == 0**.

drive(l, r)

One function for every movement: **drive(150, 150)** forward, **drive(110, -110)** spin, **drive(0, 0)** stop.

Thresholds

Put YOUR calibration numbers in SEE and CLOSE.

Objective 3 · State machine · Build step 3 of 4

Program part 2: the state machine

```
void loop() {
  int L = analogRead(A0), C = analogRead(A1), R = analogRead(A2);
  int best = min(C, min(L, R));                // smallest = strongest IR

  if (state == SEARCH) {
    drive(110, -110);                          // spin slowly
    if (best < SEE) state = APPROACH;
  }
  else if (state == APPROACH) {
    if (best > SEE)          state = SEARCH;  // lost it
    else if (C < CLOSE)      { drive(0, 0); state = EXTINGUISH; }
    else if (best == L)      drive(60, 160);  // curve left
    else if (best == R)      drive(160, 60);  // curve right
    else                     drive(150, 150); // straight on
  }
  else {                                       // EXTINGUISH
    digitalWrite(FAN, HIGH);  tone(BUZ, 1500, 100);
    for (int a = 60; a <= 120; a += 5) { sweep.write(a); delay(40); }
    for (int a = 120; a >= 60; a -= 5) { sweep.write(a); delay(40); }
    if (analogRead(A1) > SEE) { digitalWrite(FAN, LOW); state = SEARCH; }
  }
  Serial.println(state);                       // 0, 1 or 2
}
```

Spinning too fast?

It may spin past the flame. Lower 110, or stop briefly every half turn to look.

Never stuck on

When the flame goes out, the fan switches OFF. Safe defaults matter.

Check

Serial Monitor shows 0 → 1 → 2 → 0 as the robot searches, approaches and extinguishes.

Objective 5 · Build step 4 of 4

Arena trials

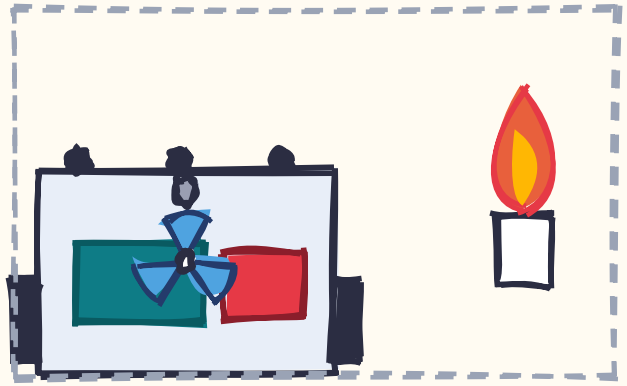
a Mark a 2 m × 2 m arena with tape. Tealight in a tray at a random spot.

b Place the robot facing **away** from the flame. Start the timer.

c Record: time to find, time to extinguish, did it stop 15 cm away?

d Run 5 trials from different starting positions.

e Then try one 'trick' trial: phone torch on one side, candle on the other.



Check: The robot finds and puts out the flame in under 30 s, stopping safely.

Robot heading for a person or bag? Lift it up (it's safe) and fix the code before the next trial.

Test like an engineer

Trial results

Trial	Find (s)	Out (s)	Stopped safely?
Trial 1			
Trial 2			
Trial 3			
Trial 4			
Trial 5			
Trick trial (torch)			

Debug checklist

Spins forever? SEE threshold too low. Crashes into the tray? CLOSE threshold too high, or sensors pointing up. Resets when the fan starts? Fan from the battery, not the Arduino 5V.

Change ONE value per trial and record it. Otherwise you won't know what helped.

Level up!

Robot challenges

Choose your challenge



Sensor fusion: add a temperature sensor; only extinguish if IR AND heat agree.



Obstacle aware: add an ultrasonic sensor so it drives round a box to reach the flame.



Multi-fire: two candles: put out both, then return to the start.



Report home: send 'FIRE FOUND' and the time to a phone (ESP32 or Bluetooth).

Think like an engineer

Why might a robot spin past the flame without noticing it? How would you fix it?

Boss level

Fastest safe time across the class wins: find + extinguish + stop at 15 cm.

Think about it

When should a robot decide alone?

Scenario

Our robot chased a phone torch, a false alarm. In a real building, a fire robot sprays water on what it thinks is fire. It is actually a heater next to a sleeping person.

Discuss

- Which decisions should always need a human?
- How could the robot **double-check** (two kinds of sensor)?
- Who is responsible for its mistakes?

Engineering idea

Sensor fusion: combine IR with a temperature sensor or camera, so one fooled sensor can't trigger action.

Objective check

Can I really do it? Prove it.

01	Show your calibration table. What are your SEE and CLOSE values?	○ ○ ○
02	Point to the line that makes the robot curve left. Why that way round?	○ ○ ○
03	Draw the state diagram from memory. What triggers each change?	○ ○ ○
04	Show the fan sweeping. What happens when the flame goes out?	○ ○ ○
05	Read your trial results. What false alarm did you find, and how could you prevent it?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Robot showcase: each team runs one timed trial in front of the class.

Show what you know

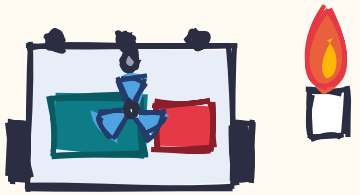
What did we learn?

- [x] Flames give off IR; sensors read lower numbers for stronger IR
- [x] Calibrate thresholds in YOUR room, with YOUR sensors
- [x] Steer by comparing left, centre and right
- [x] A state machine keeps behaviour clear: search, approach, extinguish
- [x] Safe defaults and human oversight matter for autonomous robots

Exit ticket: the robot keeps turning towards the window. What is happening, and name two fixes.

At home

Find out what kind of fire extinguishers your school has, and which fires each one is for.



Next class:
Parametric Robot Gripper Design!